

1, Explain behaviour of hub device

Does not learn MAC addresses.
Does not inspect destination MAC
sending incoming data to all other ports
Creates one collision domain.
Usually operates using half-duplex

2, Explain behaviour of switch device

→ learns source MAC addresses
→ stores MAC → Port mappings in a MAC table
→ checks destination MAC
→ Each switch port is normally a separate collision domain.
→ Broadcast traffic occurred.

3, Explain behaviour of router device

→ Uses IP addresses.
→ maintains a routing table.
→ Connects different networks.
→ Separates broadcast domains.
→ Each router interface belongs to a different network / subnet.